

LAB_7

1. Write a program that tries to access an element outside the bounds of an array and handles the `ArrayIndexOutOfBoundsException` by printing a user-friendly message.

CODE:-

```
package lab_7;

public class ExceptionHandling {

    public static void main(String[] args) {

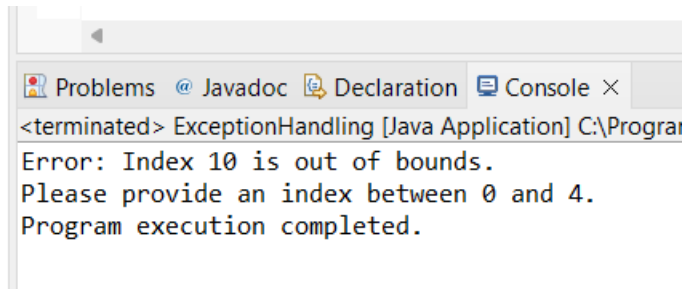
        // Define an array with 5 elements
        int[] numbers = {10,20,30,50,60};

        // Index that is out of bounds
        int index = 10;

        try {
            // Attempt to access an element at the out-of-bounds index
            int value = numbers[index];
            System.out.println("The value at index " + index + " is: " + value);
        }
        catch (ArrayIndexOutOfBoundsException e) {
            // Handle the exception with a user-friendly message
            System.out.println("Error: Index " + index + " is out of bounds.");
            System.out.println("Please provide an index between 0 and " +
(numbers.length - 1) + ".");
        }

        // Continue with the program
        System.out.println("Program execution completed.");
    }
}
```

OUTPUT:-



2. Write a program that attempts to divide a number by zero and handles the `ArithmeticException` by printing a message that division by zero is not allowed.

CODE:-

```
package lab_7;
import java.util.Scanner;
public class DivideByZero {

    public static void main(String[] args) {

        //creating the scanner object
        Scanner sc=new Scanner(System.in);

        //getting the user input
        System.out.println("enter the first number: ");
        int num=sc.nextInt();

        //getting the user input
        System.out.println("enter the second number: ");
        int num1=sc.nextInt();

        //for dividing two number
        int result= num/num1;

        //for printing two result
        System.out.println("Result:"+ result);

        //closing the scanner object
        sc.close();
    }
}
```

OUTPUT:-

```
Problems  @ Javadoc  Declaration  Console x
<terminated> DivideByZero [Java Application] C:\Program Files\Java\jdk-21\bin\javaw.exe (Aug 1
enter the first number:
10
enter the second number:
0
Exception in thread "main" java.lang.ArithmeticException: / by zero
at ClassAnudip/lab_7.DivideByZero.main(DivideByZero.java:20)
```

3. Write a Java program that reads an integer input from the user and throws an `IllegalArgumentException` if the input is negative. Display an appropriate message when the exception is caught.

CODE:-

```
package lab_7;
import java.util.Scanner;
public class ExceptionArgu {

    public static void main(String[] args) {

        //creating the scanner object
        Scanner sc=new Scanner (System.in);

        try {

            //getting input from the user
            System.out.println("enter the number: ");
            int num=sc.nextInt();

            //checking that user is entering the negative number or not if yes it will
            throw a IllegalArgumentException
            if(num<0) {
                throw new IllegalArgumentException ("negative numbers are
                not allowed:"+num);
            }
            else {
                System.out.println("enter number is positive: "+num);
            }
        }
        catch (IllegalArgumentException e){
            System.out.println("Exception caught: "+e.getMessage());
        }
        finally {
```

```

        //closing the scanner object
        sc.close();
    }
}

```

OUTPUT:-

```

<terminated> ExceptionArgu [Java Application] C:\Program Files\Java\jdk-21\bin\java.exe -Djava.class.path=.;C:\Program Files\Java\jdk-21\bin\java.exe -jar ExceptionArgu.jar
enter the number:
-9
Exception caught: negative numbers are not allowed:-9

```

4. Create a Java method that divides two numbers and declares that it throws an `ArithmeticException`. Handle the exception in the main method.

CODE:-

```
package exceptionDemo;
```

```
public class DivisionExceptionDemo {
```

```

    // for divide two numbers, declaring it throws an ArithmeticException
    public static int divide(int num, int num2) throws ArithmeticException
    {
        return num / num2; // This will throw ArithmeticException if denominator
is 0
    }

```

```

    public static void main(String[] args)
    {
        try
        {
            //values for division method
            int result = divide(10, 0);
            System.out.println("Result: " + result);
        }
        catch (ArithmeticException e)
        {
            // Handling the ArithmeticException if it is thrown
            System.out.println("Exception caught: Division by zero is not
allowed.");
        }
    }
}

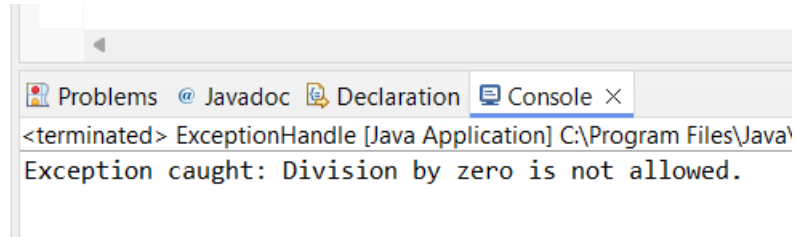
```

```

    }
}
}

```

OUTPUT:-



5. Define a custom exception called `InvalidAgeException`. Write a Java program that throws this exception if the age provided is less than 18. Handle the exception and display an appropriate message.

CODE:-

```

package lab_7;
import java.util.Scanner;
public class CustomeDemo {
    // Method to validate age, throws InvalidAgeException if age < 18
    public static void validateAge(int age) throws InvalidAgeException
    {
        if (age < 18)
        {
            throw new InvalidAgeException("Age must be 18 or older.
Provided age: " + age);
        } else
        {
            System.out.println("Age is valid: " + age);
        }
    }

    public static void main(String[] args)
    {
        // Create a Scanner object to read input from the user
        Scanner scanner = new Scanner(System.in);

        try
        {
            // Prompt the user to enter their age
            System.out.print("Enter your age: ");
            int age = scanner.nextInt();

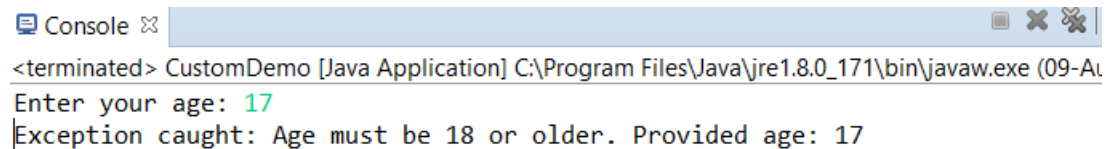
```

```

        // Validate the age using the validateAge method
        validateAge(age);
    }
    catch (InvalidAgeException e)
    {
        // Handle the InvalidAgeException and display an appropriate
message
        System.out.println("Exception caught: " + e.getMessage());
    }
    finally
    {
        // Close the scanner
        scanner.close();
    }
}
}

```

OUTPUT:-



```

<terminated> CustomDemo [Java Application] C:\Program Files\Java\jre1.8.0_171\bin\javaw.exe (09-Au
Enter your age: 17
Exception caught: Age must be 18 or older. Provided age: 17

```

-
6. Write a Java program that has a method to validate a user's email address. The method should throw a custom exception `InvalidEmailException` if the email does not contain `@` and `..`. Handle the exception in the main method.

CODE:-

```

package lab_7;
import java.util.Scanner;
class InvalidEmailException extends Exception{
    public InvalidEmailException(String message)
    { super(message);}
}
public class InvalidEmail {
    // Method to validate the email address, throws InvalidEmailException if invalid
    public static void validateEmail(String email) throws InvalidEmailException {
        // Check if the email contains "@" and "."

```

```

        if (!email.contains("@") || !email.contains(".")) {
            throw new InvalidEmailException("Invalid email address: " +
email);
        } else {
            System.out.println("Email is valid: " + email);
        }
    }

    public static void main(String[] args) {
        // Create a Scanner object to read input from the user
        Scanner scanner = new Scanner(System.in);

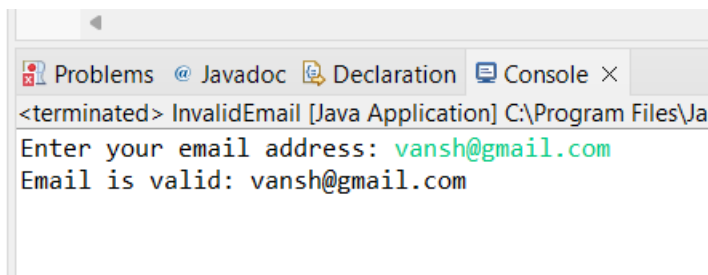
        try {
            // Prompt the user to enter their email address
            System.out.print("Enter your email address: ");
            String email = scanner.nextLine();

            // Validate the email address using the validateEmail method
            validateEmail(email);
        } catch (InvalidEmailException e) {
            // Handle the InvalidEmailException and display an appropriate
message

            System.out.println("Exception caught: " + e.getMessage());
        } finally {
            // Close the scanner
            scanner.close();
        }
    }
}

```

OUTPUT:-



```

<terminated> InvalidEmail [Java Application] C:\Program Files\Ja
Enter your email address: vansh@gmail.com
Email is valid: vansh@gmail.com

```